A Review on Social Media Based Profiling Analysis

Ema Utami
Magister of Informatics Engineering
Universitas Amikom
Yogyakarta-Indonesia
Email:ema.u@amikom.ac.id

Mihuandayani
Magister of Informatics Engineering
Universitas Amikom
Yogyakarta-Indonesia
Email:mihuandayani20@gmail.com

Suwanto Raharjo
Informatics
IST AKPRIND
Yogyakarta-Indonesia
Email: wa2n@akprind.ac.id

Anggit Dwi Hartanto
Faculty of Computer Science
Universitas Amikom
Yogyakarta-Indonesia
Email:anggit@amikom.ac.id

Sumarni Adi
Faculty of Computer Science
Universitas Amikom
Yogyakarta-Indonesia
Email:sumarni.a@amikom.ac.id

***Abstract*—Social media is one of the many internet services for users of productive age. Major social media such as Facebook, Instagram, and Twitter are having many users in Indonesia. The existence of these applications creates new impacts on social interaction. The behaviour of social media users possible to reflect the character of the user. Some cases related to a person's character often occur starting from what their activities do on social media, for example concerning the relationship between employees and where they work. Human resources play an important role in the success of a company, thus improving the quality of human resources is the main thing. One of the processes in this improvement is by recruiting selective prospective employees. The main goal of the research is doing a literature review to see whether it is possible to use social media activities as one of the factors that can be considered for employee recruitment. This research is focused on surveying the recent journal researches about profiling analysis in social media and then looking further on the methodology, objective and variables regarding personality traits. The result of this study is the social media platforms have big opportunity to be used as one of considering parameters in the employee recruitment process.**

***Index Terms*—data mining, classification, stemming, nlp**

I. Introduction

Data utilization on social media for research purposes is very widely used. This is because the openness of data is provided and contains personal and social content from account users that can be used to see needs and comments or complaints that are felt in real-time. In this study is planned to find the possibility of the use of social media data to find out a person's personality traits for the recruitment needs of new employees.

The consideration of using social media data is the need for competent human resources and seeking a good personality and following the position of employees needed. In many sectors of the company, human resources became important assets. The activities of the Human Resource Department include human resources consisting of workforce planning, recruitment and selection, and talent management, performance management and awards, employee relations, and employee service provision. One of the duties of human resource management is to support the organization in achieving its objectives. Developing and implementing a human resource strategy integrated with the business strategies of the organization is one of them [1].

Recruitment is the first process of building an organization's human capital by finding and selecting the best candidates for filling up the positions in the organization. Selection is defined as part of the recruitment process of choosing that best candidate for the vacant jobs or positions. Interpersonal, personal skills, knowledge, experiences are critical for an organization to build the human capital. Recruiting is a complex work that requires many steps and processes. Selection tests can be used to provide data related to the organization needed and having a framework for the test is very important. Employee soft skills such as think creatively, collaborate effectively and adapt quickly are important in addition to hard skills, such as technical knowledge and competencies. Having data about soft skills, such as personal attributes and behaviours from candidates are important values for an organization.

Personality is made person unique and is made up of the characteristics, patterns of thoughts, feelings, and behaviours. Psychological qualities that influence the behaviour patterns of individual characteristics in a stable and distinctive way is one definition of personality [2]. People may show variability or inconsistency in some traits, there are many factors influence it. Person's attitudes or behaviour may be influenced by their emotion. One of most popular model of emotional intelligence is personality model. The modelling system used to reflects the consistency of behavior and adaptation of the people who fill it.

The main goal from the Human Resources Department (HRD) in an organization is managing the most valuable resources in the company, employees to meet the business needs. There are many functions in the HRD to achieve the goal, including the recruitment of employees. To start the employee recruitment process, HRD advertises job postings using online and offline media, including social media. The other processes are source candidates, screen applicants, conduct

978-1-7281-9068-6/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: Univ of Calif Santa Barbara. Downloaded on May 15,2021 at 00:52:33 UTC from IEEE Xplore. Restrictions apply.

various test namely written tests, practice tests, personality tests, and interviews. There are various types of recruitment methods, employee expectations and constantly evolving technology have changed the hiring landscape. The interview or personality test is believed the only time to assess a candidate's true personality and cultural fit. Personality assessments can became the least effective in the process because of the many variables that need to be factored into hiring a person. Assessing candidates traits during an interview or personality test would be time-consuming, exhausting and expensive.

Based on research [3], the algorithm such as Support Vector Machine with Chi-Square can be used to select the relevant variables on social media to find the candidate employee using profiling analysis based on the DISC category. Social media activities are not just an expression of personal interests but it is a window to describe a person's character or personal nature. Engagement on social media such as Instagram, Facebook, Twitter, YouTube and the others become a media to interact and express feelings through status, post and reply comments. The data from social media has been used to conduct a profiling analysis of one's personality as in research [4], [5]. Profiling is a method or technique to processes personal and non-personal data to develop predictive knowledge in building a person profiles, then applied as a basis for decision [6].

There many various digital data type in social media users timeline, namely text, images, audio, and video. Social media data are large and unstructured need to be processed such as using various text mining (processing), image processing, a bag of word approaches and others before using it. The processed data can use classification techniques to group data based on the model which divides behaviour into four different dimensions, namely Dominance, Influence, Stability and Conscientiousness (DISC) that recognize a person's feelings and behaviour that are normal in interactions with their environment [7].

II. Related Work

A study that analyzes the most important issues related to Social Media Analytics and Smart Tourism (SMAST). This research about systematic literature review also offered some guidelines for future research and resulted in twenty issues based on social media and smart tourism. That paper had a two-research question about what are the current issues related to SMAST and what are the top ranking for a future research topic in SMAST. That work emphasizes that the use of data generated by users in social networks and Smart Tourism are topics of great interest for researchers in tourism, challenges, opportunities and emerging approaches in SMAST [8].

Another research about user behaviour mining on social media which discussed by combining theorems and techniques related to machine learning, social network analysis, data mining, and others. That study was performed 174 articles were selected as primary studies and classified the surveyed studies into four categories based on the focused area: users, user-generated content, the structure of the network that content spreads on it and information diffusion [9]. The research through a critical evaluation of the systematic literature review purposes future research about social media marketing. Based on the general description that the systematic literature review aims to analyze related research, develop matrix concepts, and support research practices. Other explanations relate to the topics of strategy and social trade, social media marketing, consumer behaviour, and value creation in the business model [10].

Research that provided the current state about methods and applications for spatiotemporal analysis of the social network in the Twitter platform. That study discussed the spatiotemporal information from location-based on the social networks which represent a promising and still underexplored field for the researchers to understand geographic processes and spatial relationships inside social networks. Twitter as a potential social network used in that study to review more detailed through a systematic literature review to develop a novel method in the field of GIScience and exploration of social media data by leveraging existing geographic knowledge in Twitter [11].

A study that identified 116 primary studies about social networking sites [12], find the social media's potential for innovation purposes [13], and detection method of the customer emotions on social media platforms [14], through the systematic literature review tried to find the research trends and gaps in the field with behavioural and resource perspective theoretical exploring the social media [15]. Another research that analyzed the literature about the profile of the digital news portal to discussed data veracity which has become trending in the future [16]. Important challenges also in social media analytics which processed by discovery, collection, and preparation to point the possible solution in social media data.

III. Methodology

The literature review in this research discussed the point of how to utilize social media for person profiling and analyzing based on the personalities involved in the recruitment process. This study began with finding credible literature by surveying the recent journal researches about profiling analysis in social media and then compared the methodology, the objective and find the solution to the future works. A systematic literature review summaries existing evidence, identifying gaps and directions for future research [17].

Specifically, this study aims to address the following research questions: RQ1. What are the variables that can be used for profile analysis from social media especially for prospective employees? RQ2. What are the effective method and category of personality traits can be used for hiring prospective employees?. In this research to analyze and find the solution for the future works is done by some methodologies: define the research questions; searching by keywords related to profiling analysis in social media through the credible and relevant source databases of journal articles such as IEEE Xplore, Science Direct, ACM Digital Library; selection all the journal articles by reading the abstract and then the full paper to get the relevant journal article; analysis and classification of

Authorized licensed use limited to: Univ of Calif Santa Barbara. Downloaded on May 15,2021 at 00:52:33 UTC from IEEE Xplore. Restrictions apply.

papers by categories and issues; and also integrate findings and answer research questions to find the conclusion for the future research about profiling analysis for recruiting the prospective employees.

IV. Result and Discussion

In this section, the result of analysis profiling based on the literature review will be categorized through the component to create the profiling analysis design based on social media for the recruitment process of prospective employees. They are two categorizations of processing from the literature review are found in this study, first is shown in Fig. 1, namely text processing. The other one is image processing used for personality traits prediction, shown in Fig 2.

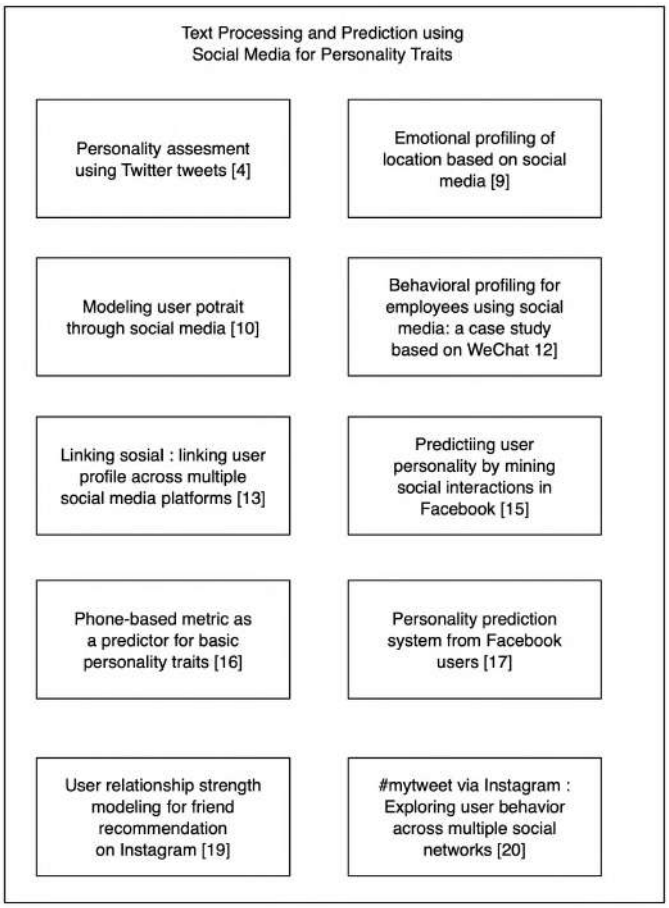


Fig. 1: Text processing in social media for personality traits

The research used several references of social media with a classification of personality traits based on DISC by managing profiles based on the classification method that carried out the comparative analysis. The research focuses on one social media platform so that there are limitations to data acquisition. While some research used various classification techniques and feature selection for general data samples to be able to represent characters in general and proposed a category based on DISC personality traits with sufficient data samples and

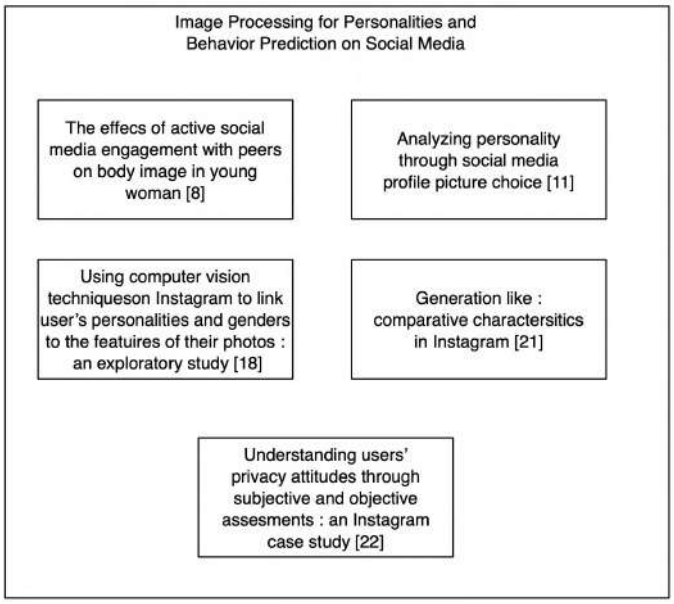


Fig. 2: Image processing in social media for personality traits

generally represented the personality of social media users to find better methods and levels of accuracy.

In some studies found the relationship that underlies the user by proposing friend recommendations using methods and experiments on Instagram. The utilization of various social media platforms that take advantage of differences in behaviour between users as part of friend recommendations and find group activities to find personality traits. In another study discusses user activities in detecting the age of users of social media accounts while this study proposes additional insights on how to assume various ages in using social media platforms. The need for privacy information uses personal photos on social media accounts and then the privacy of respondents who access Instagram on profiles and posts needs to be accurately measured.

A detailed comparison of the analysis in several studies used as a review in profiling analysis based on social media in theory for reference in prospective recruitment employees. Through deep understanding related to the relationship between the paper referred to the suitability of profiling content to be built, the analysis through the platform, how the algorithm works, and the opportunities of each referenced paper are used as evaluation material to create a profiling system in future studies. A detailed comparison of the analysis is shown in Table I.

In a company, there are several stages of selection carried out to find the right prospective employees to occupy the required positions. The selection process usually consists of administrative selection, written tests, practical tests, psychological tests, health tests, and interview tests. In one of the selection processes namely psychological tests, profiling analysis plays a role in supporting the prospective employee selection stage. Through a statistical approach to determine the

Authorized licensed use limited to: Univ of Calif Santa Barbara. Downloaded on May 15,2021 at 00:52:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I: The comparison analysis of review process

Paper Reference	The Objective of the paper	Conclusion	Suggestion/Weakness
Ref. [4]	The research aims to analyze how Twitter as a dataset can be used to improve user experience with a personality assessment. The study estimated the personality traits of a set of behavioral variables extracted. Through assessment of DISC, this study proposes ways to predict user personality that are useful in career development by mapping information through Twitter data	The framework in the research that uses text mining can predict personality traits well on Twitter data. The study found that only extraversion could be predicted to be significantly better at 35.6% than by null models.	The study needs to add several parameters to expand the research findings such as geolocation and trends.
Ref. [18]	A study that addresses the effects of involvement in social media with women's body image. Analysis of mixed variance shows a significant interaction between conditions and time that the experiments on Facebook and Instagram platforms. The research was carried out by determining the analog scale of visual images of the body and commenting on social media sites on student participants.	Studies that show active involvement with peers on social media cause a deteriorating body image in young adult women. ASME with families is not more attractive than oneself does not cause changes in body image. This study shows this activity can cause a slight increase in negative body image.	The sample used was only female students from one university, so it was not generalized to all young adult women and not to generalize men. We need to pay attention to whether ASME with peers influences the body image of young men and if certain individuals are more influenced by ASME with peers.
Ref. [19]	This research utilizes social media for the method of creating an emotional profile of a location-based on the social analysis of media interactions that have been geographically located in the long-term emotional rating of the interaction of Twitter as a social media platform. The method used is based on the multivariate kernel density function of the social media interaction by validating the approach to the location in real life.	The proposed method uses a component of emotional state models, an extended version of effective norms for English-language words for emotional extraction from social media users. And the evolution of the model to map scores to one emotional condition.	Adopting a user-centric approach such as creating a user's emotional profile over a longer period time, then mapped to the location becomes more consistent or considers segmentation by sex and education class that represents the entire population of the location.
Ref. [20]	The research that proposes psychological modeling methods based on linguistic computational features to the profile of the user personality traits on Twitter and their correlation with the user's social behavior. The study uses the big five profile characterizes personality traits namely Openness, Conscientiousness, Extroversion, Agreeableness, and Neuroticism.	This modeling approach is effective for understanding portraits of the demographics and psychology of users on social media without customer interference, which is useful for commercial merging to provide better personalization of products and services. The results of the calculation analysis show that the linguistic style, the description tag itself, the use of emoticons and demographic information can be used to calculate personality profiles without customer disruption.	Based on a trained computational psychology model, it can use more clues about the relationship between the user's personality profile and music tastes and shopping habits that can be found.
Ref. [21]	Research that analyzes Twitter profile images is different from the personality of users who post them. The images used to focus on the analysis of aesthetics and facial features and controls for demographic variations in features and personality images. Using big five personality traits from social media by finding a correlation with personality and then compared to a standard psychological questionnaire.	The results of the study showed a significant difference in the choice of profile images between personality traits with accurate accuracy. Each personality trait has a characteristic type of profile picture post. Users who are high in openness or neuroticism post fewer photos of people and tend to express positive emotions. Users who have a conscience, conformity or extraversion prefer images and positive emotions on their faces.	We need to consider the experiments in analyzing social media profile images using features that are interpreted on a larger data set. Analyze a more diverse set of psychological traits by looking at a wider collection of photos that are made by users, curated involved with using social media.
Ref. [22]	This research is used to characterize employee behavior patterns in a variety of automatic and accurate ways for organizational management. Validation is used based on real-world datasets. This research utilizes social media platforms such as WeChat and Microblog by proposing a profiling behavior approach to better analyze and understand employees, by inputting several real-time data from employees' daily work reported on social media.	WeChat's robot platform is designed to gather the daily work of employee behavior data, transfer raw data into structured records and get key features through data analysis. Then build a skill knowledge graph to formulate the PO fit model to do empirical learning and validate the model and analysis using real-world datasets released by the organization.	Analyze employee behavior using LSTM's Long Model Short Term Memory to build a personalized behavior model for each employee. Besides that, it needs to increase the Wechat robot platform with blockchain.

Continue on the next page

Authorized licensed use limited to: Univ of Calif Santa Barbara. Downloaded on May 15,2021 at 00:52:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I: The comparison analysis of review process (cont.)

Ref. [23]	This research focuses on linking on several link social media platforms with several steps, namely user profile feature extraction, feature weight calculation, and multi-platform linking from user profiles. The links are used from three platforms, namely Google+, Instagram and Twitter. Candidate grouping profiles are based on trimming different profile comparisons.	The results of the research about grouping social links for experiment selection show social link predictions with 92% accuracy in pair-wise and 74% in multi-platforms connecting three social media platforms. Linkage user profiles (UPL) across social media platforms provide a foundation for understanding users' online lives.	Extending user behavior across platforms that do not yet exist, adding link social features using heterogeneous data including text, video and images, network similarity, and patterns across social media platforms. And link social on more social media platforms.
Ref. [24]	Research is used to find patterns of general behavior when interacting through virtual social networks, and these patterns are predicted for personality trends. Facebook user interaction analysis is used to gather information about the personality and predict based on experiments 3-class models and 5-class models using machine learning techniques.	The proposed classification method used produces a high level of accuracy in making user predictions on parameters related to user interactions based on many friends or number of posts. The study shows that Facebook profile data developed with TP2010 can be used to predict user personality with the accuracy of the higher classifier than 70% achieved for all personality traits.	Restrictions on the number of Facebook database queries are an obstacle when extracting information about one user directly for example for some users who have more than a thousand friends.
Ref. [25]	The study investigated the extent to which the predictability of the use of smartphones for personality character predictions was based on the Big Five Personality Traits, namely Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism using SVM with a radial base kernel function for the classification model.	The research resulted from a set of behaviors extracted variables found that only extraversion personality traits could be predicted to be significantly better than 0 models.	The study used a sample of student data at the Technical University of Denmark where the findings could not necessarily be generalized to the population in general.
Ref. [26]	The research proposes building a system that can predict a person's personality based on Facebook user information using 5 major personality traits. This study used the machine learning algorithm and deep learning.	The study produced a comparison by conducting a comprehensive analysis method through the results of accuracy obtained an average of 74.17%. The method of selecting features can affect the performance given by the classification algorithm. The combination of selection feature classification on the data produces a suitable combination such as chi-square and information gain through approximately equivalent groups.	In experiments conducted showing machine learning used despite increasing accuracy but accuracy is still quite low for some properties because some small datasets are used in the research.
Ref. [27]	The study was conducted to determine the relationship between the characteristics of Instagram users and the photo features of account users. In this study, the correlation was observed from several features and extravagancies and was predicted based on the Big Five Personality Traits.	The study produced content categories that were associated with user extraversion and gender. While Extraversion, the openness of the user is partly related to the emotions seen in the photo collection of the face of the user account.	The limitations of the data sample used at two university campuses in South Korea in generalizing the results. Subsequent research can use samples with wider coverage, for example from different countries.
Ref. [28]	The study proposes a framework for discovering the strengths of social relationships that underlie activities and friendship recommendations on Instagram. The method used the recommendations of friends on Instagram, especially when the number of recommendations is more than 20.	The study shows that the effectiveness and work efficiency proposed in the experiment that gets better performance uses coarse to fine methods than other related methods. The proposed framework can be used for other social media communities.	The stages of normalizing the matrix of similarities between other users can eliminate some of the effects of different behaviors. Furthermore, the use of differences in behavior between users to make friend recommendations is still a major problem.
Ref. [29]	The study studied a variety of online social networks to analyze the temporal and topical behavior of various users that showed different behavior patterns on different networks. The research examined the user actions in the social media timeline or its activities through several other social networking sites simultaneously.	Research produces varied behaviors on different social media, finding dichotomies between uses for professional and personal reasons. The study validates the hypothesis that users show varying behavior on different social networking sites. Analysis of the characteristics of adolescents on social media relies heavily on ethnographic accounts or quantitative analysis in small datasets.	The study examined various social networking sites from users' public activities by not further exploring the differentiation between the use of public and private social media.
Ref. [30]	The research detected age information on user profiles by using a combination of methods of text and face recognition on Instagram.	The study found a tendency for teens to post a few photos and many tags to receive many comments and likes. Then posting more self-expression which shows a higher representation than adults. The results of these studies are demonstrated from applications that show trends in age differences on social media.	The research allows for the detection of age information from the site used. It is necessary to expand social media platforms and validate user age information by comparing most selfie users to account profile photos.
Ref. [31]	The study investigated the understanding of user attitudes towards 18 items of privacy and attitude changes. The research identified through comparative analysis by discussing the implications of the results to support managing user privacy on social media.	The study found that 18 privacy items represented many sides of social media users, but more categories of personal information that did not include research were on data obtained from Instagram.	In conducting learning techniques to extract knowledge from images, more privacy is needed regarding the information. The study has several limitations such as research respondents still in a hypothetical scenario.

Authorized licensed use limited to: Univ of Calif Santa Barbara. Downloaded on May 15,2021 at 00:52:33 UTC from IEEE Xplore. Restrictions apply.

variables used to find someone's profile, the design is precisely described and modelled so that a profiling analysis can be built through the application of algorithms. The process of profiling starts with preparing a dataset from personal pages on social media platforms. In the profiling process, it is necessary to identify data types such as status updates and comments on the user's page.

The profiling process can use the Application Programming Interface (API) to get authentication keys and store data in a format. After getting the data from the personal page, the process of cleaning the data becomes an important part to ensure the data is given a function to replace the missing value, set roles, nominal to text, change links, replace the hashtag symbol and then delete duplicate data. This process is also called the preprocessing stage, the process for unstructured data to be classified correctly. The main function of preprocessing text is identified words in the basic unit of analysis and tagging components and is part of a natural language processing system [32].

Determining class labels into some of the personality categorizations such as DISC is a main important part of this profiling method. Corpus-based theory must be carried out to do this labelling process. How to identify the word from a personal status update or comment from into the DISC style need the right dictionary and need an expert to construct it. What words that can identify a type of Dominance, Influence, Stability or Conscientiousness can be built by help from feature selection method.

These results are then classified with the right algorithm so that the results of data validity and algorithm measurement can be tested. The implementation of profiling analysis that is applied is used to evaluate and follow up so that profiling optimization can be utilized effectively with factual or proven data. The results of profiling analysis can be used by companies to place prospective employees appropriately based on alternative personality elements obtained through social media activities. The systematic literature review of any kind of paper related to the profiling analysis done by identifies the platform of the social media users and how the process with the parameter of the paper used to find the linkages of the user's personality.

V. CONCLUSION

This study analyzes social media activities as one of the factors that can be taken into consideration in hiring employees. Through a literature review, we can analyze and find out social media platforms can be used as well as the parameters used for profile analysis. Renewal of the social media data timeline can be used to analyze prospective employee personality profiles based on personality trait parameters. The clustering or classification of social media users is done in personality traits.

Based on the results of the analysis of several literature references used in this study, the variables used can be status updates, comments or likes, sharing posts, or joining groups are examples of variables to be examined. This variable is used to find the personality categories of prospective employees in the recruitment process. The use of social media platforms in the recruitment process needs to be adjusted on Facebook, Twitter, Instagram, and others. Furthermore, the study has been categorized the main concept of the profiling analysis for the processing data into text processing and prediction using social media for personality traits, and image processing for personalities prediction and behaviour on social media. The appropriate categorization personality model must be planned for research to conduct profiling in supporting the company's recruitment process.

ACKNOWLEDGMENT

The authors would like to acknowledge valuable funding provided by the Ministry of Research, Technology, and Higher Education of the Republic of Indonesia in Hibah Penelitian Terapan Unggulan Perguruan Tinggi (PTUPT) of the fiscal year 2019–2021, Contract Number: 227/SP2H/LT/DRPM/2019. This research also can be done by support and facility from Universitas Amikom Yogyakarta.

REFERENCES

[1] M. Armstrong, *A handbook of human resource management practice*. Kogan Page Publishers, 2006.
[2] D. A. Buchanan and A. A. Huczynski, *Organizational behaviour*. Pearson UK, 2019.
[3] Mihuandayani, E. Utami, and E. T. Luthfi, "Profiling analysis based on social media for prospective employees recruitment using SVM and chi-square," *Journal of Physics: Conference Series*, vol. 1140, p. 012043, dec 2018.
[4] N. Ahmad and J. Siddique, "Personality assessment using twitter tweets," *Procedia computer science*, vol. 112, pp. 1964–1973, 2017.
[5] G. Farnadi, J. Tang, M. De Cock, and M.-F. Moens, "User profiling through deep multimodal fusion," in *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*, 2018, pp. 171–179.
[6] V. Ferraris, F. Bosco, G. Cafiero, E. D'Angelo, and Y. Suloyeva, "Defining profiling," *Available at SSRN 2366564*, 2013.
[7] W. M. Marston, *Emotions of normal people*. Routledge, 2013, vol. 158.
[8] M.-S. Viñan-Ludeña, "A systematic literature review on social media analytics and smart tourism," in *Smart Tourism as a Driver for Culture and Sustainability*. Springer, 2019, pp. 357–374.
[9] R. M. Safari, A. M. Rahmani, and S. H. Alizadeh, "User behavior mining on social media: a systematic literature review," *Multimedia Tools and Applications*, vol. 78, no. 23, pp. 33 747–33 804, 2019.
[10] J. Rowley and B. Keegan, "An overview of systematic literature reviews in social media marketing," *Journal of Information Science*, p. 0165551519866544, 2019.
[11] E. Steiger, J. P. De Albuquerque, and A. Zipf, "An advanced systematic literature review on spatiotemporal analyses of t witter data," *Transactions in GIS*, vol. 19, no. 6, pp. 809–834, 2015.
[12] H. Waheed, M. Anjum, M. Rehman, and A. Khawaja, "Investigation of user behavior on social networking sites," *PloS one*, vol. 12, no. 2, 2017.
[13] H. Bhimani, A.-L. Mention, and P.-J. Barlatier, "Social media and innovation: A systematic literature review and future research directions," *Technological Forecasting and Social Change*, vol. 144, pp. 251–269, 2019.
[14] P. Madhala, J. Jussila, H. Aramo-Immonen, and A. Suominen, "Systematic literature review on customer emotions in social media," in *ECSM 2018 5th European Conference on Social Media*. Academic Conferences and publishing limited, 2018, p. 154.
[15] S. Affendey and A. Mamat, "A systematic review on the profiling of digital news portal for big data veracity," *Procedia Computer Science*, vol. 72, pp. 390–397, 2015.
[16] S. Stieglitz, M. Mirbabaie, B. Ross, and C. Neuberger, "Social media analytics–challenges in topic discovery, data collection, and data preparation," *International journal of information management*, vol. 39, pp. 156–168, 2018.

Authorized licensed use limited to: Univ of Calif Santa Barbara. Downloaded on May 15,2021 at 00:52:33 UTC from IEEE Xplore. Restrictions apply.

[17] M. Petticrew and H. Roberts, "How to appraise the studies: an introduction to assessing study quality," *Systematic reviews in the social sciences: A practical guide*, pp. 125–163, 2006.
[18] J. V. Hogue and J. S. Mills, "The effects of active social media engagement with peers on body image in young women," *Body image*, vol. 28, pp. 1–5, 2019.
[19] J. Bernabé-Moreno, A. Tejeda-Lorente, C. Porcel, H. Fujita, and E. Herrera-Viedma, "Emotional profiling of locations based on social media," *Procedia Computer Science*, vol. 55, pp. 960–969, 2015.
[20] H. Gu, J. Wang, Z. Wang, B. Zhuang, and F. Su, "Modeling of user portrait through social media," in *2018 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 2018, pp. 1–6.
[21] L. Liu, D. Preotiuc-Pietro, Z. R. Samani, M. E. Moghaddam, and L. Ungar, "Analyzing personality through social media profile picture choice," in *Tenth international AAAI conference on web and social media*, 2016.
[22] X. Ni, S. Zeng, R. Qin, J. Li, Y. Yuan, and F.-Y. Wang, "Behavioral profiling for employees using social media: A case study based on wechat," in *2017 Chinese Automation Congress (CAC)*. IEEE, 2017, pp. 7725–7730.
[23] V. Sharma and C. Dyreson, "Linksocial: linking user profiles across multiple social media platforms," in *2018 IEEE International Conference on Big Knowledge (ICBK)*. IEEE, 2018, pp. 260–267.
[24] A. Ortigosa, R. M. Carro, and J. I. Quiroga, "Predicting user personality by mining social interactions in facebook," *Journal of computer and System Sciences*, vol. 80, no. 1, pp. 57–71, 2014.
[25] B. Mønsted, A. Mollgaard, and J. Mathiesen, "Phone-based metric as a predictor for basic personality traits," *Journal of Research in Personality*, vol. 74, pp. 16–22, 2018.
[26] T. Tandera, D. Suhartono, R. Wongso, Y. L. Prasetio *et al.*, "Personality prediction system from facebook users," *Procedia computer science*, vol. 116, pp. 604–611, 2017.
[27] Y. Kim and J. H. Kim, "Using computer vision techniques on instagram to link users' personalities and genders to the features of their photos: An exploratory study," *Information Processing & Management*, vol. 54, no. 6, pp. 1101–1114, 2018.
[28] D. Guo, J. Xu, J. Zhang, M. Xu, Y. Cui, and X. He, "User relationship strength modeling for friend recommendation on instagram," *Neurocomputing*, vol. 239, pp. 9–18, 2017.
[29] B. H. Lim, D. Lu, T. Chen, and M.-Y. Kan, "# mytweet via instagram: Exploring user behaviour across multiple social networks," in *2015 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. IEEE, 2015, pp. 113–120.
[30] J. Y. Jang, K. Han, P. C. Shih, and D. Lee, "Generation like: Comparative characteristics in instagram," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2015, pp. 4039–4042.
[31] K. Han, H. Jung, J. Y. Jang, and D. Lee, "Understanding users' privacy attitudes through subjective and objective assessments: An instagram case study," *Computer*, vol. 51, no. 6, pp. 18–28, 2018.
[32] N. Indurkhya and F. J. Damerau, *Handbook of natural language processing*. Chapman and Hall/CRC, 2010.

Authorized licensed use limited to: Univ of Calif Santa Barbara. Downloaded on May 15,2021 at 00:52:33 UTC from IEEE Xplore. Restrictions apply.